



**THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MBINGA DISTRICT COUNCIL
FORM IV DISTRICT PRE MOCK EXAMINATIONS MARCH 2026**



031

PHYSICS

Time: 3:00 HOURS

Thursday 19th March 2026 A.m.

INSTRUCTIONS

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and **TWO (2)** question from section C.
3. Non programmable Calculators is allowed in the examination room.
4. Write your **Examination Number** on every page of your answer booklet(s).
5. Where necessary the following constants may be used:
 - (i) Acceleration due to gravity, $g = 10 \text{ m/s}^2$ or 10 N/kg
 - (ii) Specific heat capacity of mercury is $1395 \text{ J/kg}^\circ\text{C}$
 - (iii) Density of water, $= 1 \text{ g/cm}^3$ or 1000 kg/m^3

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SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) - (x), choose the correct answer among the given alternatives and write its letter besides the item number in the answer booklet provided.
 - i Which pairs of instruments would you use to correctly measure the diameter of a small ball bearing?
 - A. Measuring tape and vernier caliper
 - B. Slide rule and micrometer screw vernier caliper
 - C. Vernier caliper and slide rule
 - D. Micrometer screw gauge and
 - E. Metre rule and micrometer screw gauge
 - ii. Why is oil used as a lubricant?
 - A. has low density
 - B. it is flammable
 - C. it is inflammable
 - D. is high viscous
 - E. it is less viscous.

iii. Which one is a characteristic of a plane mirror?

- A. It forms an image which is real and opaque
- B. It forms an image which is larger than the object
- C. It forms an image which is real and laterally inverted
- D. It forms an image which has the same size as the object
- E. It forms an image which is small in size than the object

iv. Which one is **not** a region of electromagnetic spectrum?

- A. radio waves
- B. x- rays
- C. infrared light
- D. ultraviolet light
- E. invisible light waves

v. What quantity of heat is required to raise the temperature of 25 kg sample of mercury from 20°C to 30°C?

- A. 1,743,750 J
- B. 348,750 J
- C. 345,750 J
- D. 1,550,750 J
- E. 413,750 J

vi. When silicon element is doped with phosphorus atoms, it produces

- A. a p-type semiconductor
- B. a p-n junction diode
- C. an intrinsic semiconductor
- D. a bipolar semiconductor
- E. an n-type semiconductor

vii. Which of the following is an example of a scalar quantity?

- A. Electric current
- B. Force
- C. Velocity
- D. Displacement
- E. Acceleration

viii. What role does the iris play in the human eye?

- A. To hold the lens in position
- B. To prevent internal reflection
- C. To control the size of the pupil
- D. To control the thickness of the lens
- E. To protect the eye from light

ix. Asteroids which manage to reach the earth surface are called

- A. stars
- B. Meteors
- C. Meteorites
- D. Constellations
- E. Comets

x. What will be the resistivity of a wire 2 metres long with a cross-sectional area of 0.50 mm² and a resistance of 2.20 Ω

- A. $5.5 \times 10^{-7} \Omega \text{m}$
- B. $6.5 \times 10^{-7} \Omega \text{m}$

- C. $2.3 \times 10^{-7} \Omega \text{m}$ E. $5.5 \times 10^{-6} \Omega \text{m}$
 D. $1.1 \times 10^{-6} \Omega \text{m}$

2. Match the functions of the features of cathode ray tube in **List A** with the corresponding features in **List B** by writing the letter of the correct response beside the item number in the answer booklet provided. (05 marks)

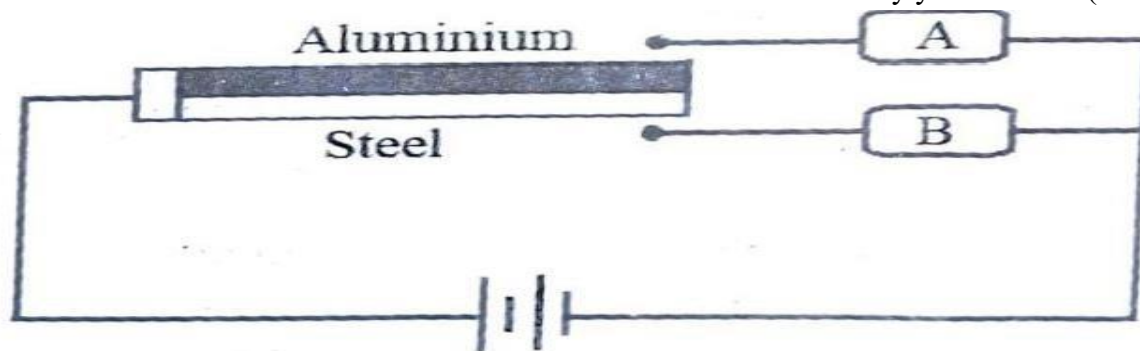
List A	List B
i Materials that can strongly be magnetized. ii Substance which are made up of soft iron. iii Materials that cannot be affected by magnets. iv Objects which are made up of steel. v Groups of magnetic dipoles arranged themselves in a magnetized object. vi Field lines of force used for finding locations of different places.	A. Paramagnetic B. Temporary magnets C. Zinc and Copper D. Permanent magnets E. Magnetic domains F. Earth's magnetic field G. Induced magnetism H. Magnetization I. Demagnetization J. Permeable K. Neutral point L. Ferromagnetic M. Magnetic field.

SECTION B (54 Marks) Answer **all** questions in this section.

- 3.(a) In a light experiment, a narrow beam of light directed onto a glass prism leaves the prism and falls on a white screen. Draw a labelled diagram to show the experimental set-up and observation seen on a screen. **(4.5 marks)**
- (b) Explain two ways in which lens cameras differ from the human eye. **(4.5 marks)**
- 4.(a) Why a bubble of air increases in volume as it rises from the bottom of a pond of water to the surface? Briefly explain. **(4.5 marks)**
- (b) A half meter rule AB is freely pivoted at 18 cm from end A and balances horizontally when a body of mass 35 g is hung 48 cm from end B. Calculate the mass of the rule. **(4.5 Marks)**
- 5.(a) Why the recoil velocity of a gun is much less than the velocity of the bullet? Explain by using the principle of conservation of linear momentum. **(4 marks)**
- (b) A hydraulic press consists of a pump, load and two cylinders in which the larger cylinder is eight times the diameter of a smaller one. Use a well labelled cross-sectional diagram of a

hydraulic press to determine the value of mechanical advantage of this machine if it is working with efficiency of 90%. **(4.5 marks)**

- 6.(a) Figure below shows a bimetallic thermostat strip used to regulate the temperature of the room by using a heater and a cooler. Which device between A and B is a heater? Justify your answer. (4 marks)

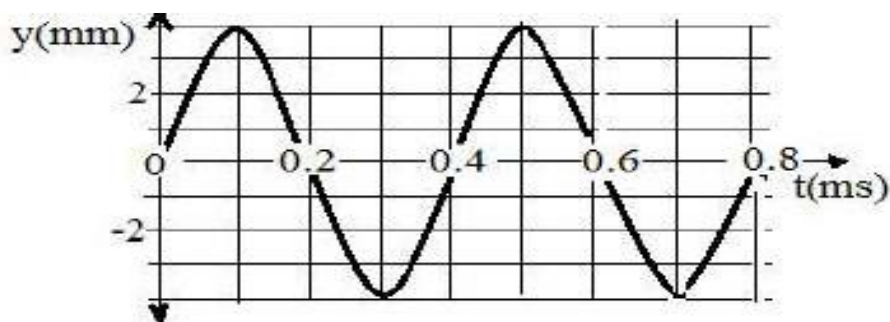


- (b) Demonstrate by using a diagram on how heat transfer by convection takes place when water is heated? (5 marks)
- 7.(a) A closed pipe which is opened at one end has a fundamental frequency of 400 Hz. Draw its mode of vibration of air column at its fundamental mode and first overtone. (4 marks)
- (b) A uranium nucleus ${}^{238}_{92}\text{U}$ decays into a lead nucleus ${}^{206}_{82}\text{Pb}$. Calculate the number of alpha and beta particles emitted.
- 8.(a) Why are the surface waves more dangerous than the primary and secondary waves? (3marks)
- (b) Describe the three uses of constellations. (6 marks)

SECTION C (30 Marks)

Answer two (2) questions from this section.

- 9.(a) A piano wire and a tuning fork produce different notes at the same time and beats are heard. What could be done on the piano wire in order to emit a note of the same frequency as vibrating fork. (Give three points) (4 marks)
- (b) A graph below shows a wave emitted by the electromagnetic source. If the velocity of the wave is 12m/s, determine the frequency and the wavelength of the wave. (6 marks)



- (c) water in the seas or lakes. A sonar pulse sent out by a boat arrives back after 3seconds. If the speed of sound in water is 1500m/s, how deep is the water? (5 marks)

- 10.(a) (i) Give the difference between conductor, semiconductor and insulator using energy band concept (3 Marks)

- (ii) Explain the mechanism of doping and give the significance of doping **(2 marks)**
- (iii) With the help of diagram distinguish transistor from diode. **(3 marks)**

- (b) Draw the circuit diagram of full wave rectifier using only two diodes. **(4marks)**
- (c) Outline any two practical applications of electronics in our daily life **(3marks)**

- 11.(a) A small house with mains supply of 240 V has two heaters each rated at 2.5 kW. The supply has a main fuse rated at 15 A. Show whether this fuse can support the operation of the two heaters at the same time. **(5 marks)**

- (b) A simple a.c generator produces a low e.m.f. Explain any three design alterations that you would make to the generator in order to convert it into one of higher e.m.f. **(6 marks)**

- (c) You are given a chance of buying and electronic device that can be used to convert the generator in part 11(b) into one of higher e.m.f. Describe one device which you will choose to buy for the purpose. **(4 marks)**